

Saturated Superfluid Vacuum VIII

Cosmogony from the Permissive Void

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Draft

Abstract

This paper proposes a staged cosmogony for the emergence of the saturated vacuum in the Saturated Superfluid Vacuum (SSV) framework. The central claim is that the present saturated, defect-bearing vacuum can be described as the end state of a growth process beginning from the permissive void, passing through primordial perturbation, interference, saturation, and topological defect nucleation.

The paper is explicitly interpretive and ontological in scope. It does not claim to present a complete quantitative cosmology. Its purpose is narrower: to state the founding logic of the cosmogonic picture clearly and separately from the quantum and gravity derivation papers. The Void is treated not as a productive mechanism, but as the absence of structure and therefore as a permissive condition that cannot forbid emergence. From this starting point, the paper distinguishes primordial time from proper time, treats the earliest universe as a gestational phase of radiative quanta of the forming medium, and interprets the Big Bang as the global release of previously stored structure rather than as the first appearance of order.

1 Introduction

This paper presents the staged cosmogony of the Saturated Superfluid Vacuum framework in its own right. The goal is not to derive every later structure of the series again, but to separate the ontological and cosmological sequence from the quantum and gravity derivations: permissive void, primordial perturbation, interference, saturation, and defect nucleation.

The body below is adapted from the legacy unified SSV V manuscript and reorganised to fit the scoped paper plan of the repository.

2 Cosmogony: From the Permissive Void to the Resonant Cosmos

We now ask the deepest question the framework permits: *why is there a saturated superfluid at all?* The answer proceeds in five stages, each following necessarily from the one before.

2.1 Stage 1 The Permissive Void

Before the superfluid existed, there was neither matter, nor energy, nor time. We posit a *permissive void*: a state of absolute zero density ($\rho = 0$), zero temperature, zero entropy, and zero curvature. It has no properties save one it does not forbid change. This is a minimal ontological assumption: the void is not nothing (a logical nullity), but rather a *substrate of pure*

potentiality a mathematical space in which the field equation (??) is defined but with $\rho = 0$ everywhere.

In this state the Madelung equations reduce trivially to $\partial_t \Psi = 0$: the void is static. But the void has an instability. The zero-density state $\Psi = 0$ is an *unstable fixed point* of the GPE whenever the interaction coefficient $g < 0$ (attractive case) or when the vacuum energy P_0 in the full equation of state provides a positive pressure at finite ρ . Either condition means that the trivial solution is not the global energy minimum the true ground state is the saturated superfluid at $\rho = \rho_0 > 0$.

2.2 Stage 2 The Primordial Photon: First Perturbation

A spontaneous, arbitrarily small transverse perturbation $\delta\Psi$ to the void state is *not forbidden*. Since the void has no thermodynamic equilibrium (it has no temperature), there is no Boltzmann suppression of fluctuations. The perturbation does not require an external cause; it is permitted by the field equation. We identify this first perturbation with the *primordial photon* a transverse phonon mode propagating through the otherwise empty void at phase velocity c (the limiting speed of transverse perturbations in the medium, as set by the ratio $\alpha c \rightarrow c$ as $\rho \rightarrow 0$).

The primordial photon is not a quantum mechanical object in the usual sense there is no pre-existing Hilbert space for it to live in. It is a classical perturbation in the nascent field Ψ , and it is the *first event*: the origin of time. Time begins not with a singularity but with a gradient the irreversible propagation of a density wave through a medium that, for the first time, is no longer uniform.

2.3 Stage 3 Interference and the Seed Pattern

A single photon propagating through the void will, on reflection or scattering from the effective boundary conditions imposed by the medium's own self-consistent density (a back-reaction), interfere with itself. Two counter-propagating photon modes of wavenumber k produce a standing wave:

$$\delta\rho(\mathbf{x}, t) = \delta\rho_0 \cos(kx) \cos(\omega t), \quad (1)$$

with $\omega = ck$ in the dilute limit. This standing wave is the *seed pattern*: a spatially modulated template of alternating high- and low-density regions. Crucially, this pattern carries *geometric information* a preferred length scale $\lambda = 2\pi/k$ from which all subsequent structure will grow.

2.4 Stage 4 Nonlinear Growth to Saturation

Where the seed pattern is constructive ($\delta\rho > 0$), the local density exceeds the void value and the GPE nonlinearity amplifies the perturbation. Where it is destructive, density remains near zero. The amplification is self-accelerating: higher local density increases the interaction energy $g\rho$, which deepens the potential well and attracts more density from neighbouring regions.

The growth proceeds through three sub-stages:

1. **Exponential amplification** (linear instability regime): the high-density crests grow as $\rho(t) \propto e^{\gamma t}$ with growth rate $\gamma = c/\xi_0$ (the inverse of the initial healing length).
2. **Nonlinear broadening**: as ρ approaches ρ_0 , the equation of state stiffens and the growth slows. The density profile develops a plateau at ρ_0 separated by sharp domain walls precursors of the vacuum surface tension that will become the strong force (Paper II).

3. **Saturation:** when $\rho \rightarrow \rho_0$ throughout the high-density crests, the medium has reached its ground state locally. The process is exothermic: the binding energy released goes into kinetic energy of the flow field, driving turbulence.

This stage corresponds, in cosmological language, to *inflation*: an exponential expansion of ordered vacuum from the initial seed, followed by a transition to the saturated state (analogous to reheating). No inflaton field is needed; the growth is driven by the superfluid’s own equation of state.

2.5 Stage 5 Vortex Nucleation: The First Particles

When neighbouring saturation fronts collide, the phase φ of Ψ must reconnect continuously across the interface. For generic phase mismatches, the only topologically consistent solution is the nucleation of a vortex filament at the intersection point – a line defect across which the phase winds by $2\pi n$.

These are the *first particles*: toroidal vortex rings (electrons) and spherical breather modes (protons) nucleating spontaneously wherever the phase mosaic of the saturating vacuum forced a topological mismatch. Their number density is set by the Kibble–Zurek mechanism:

$$n_{\text{defect}} \sim \frac{1}{\xi^3} \cdot \left(\frac{\xi}{\hat{\xi}} \right)^{3\nu/(1+\nu)}, \quad (2)$$

where $\hat{\xi}$ is the correlation length at the onset of the phase transition and ν is the correlation-length exponent of the GPE universality class ($\nu = 1/2$ for mean-field, $\nu \approx 0.67$ for the 3D XY fixed point).

The baryon-to-photon ratio $\eta \sim 6 \times 10^{-10}$ is, in this picture, not a free parameter but the Kibble–Zurek defect density at the saturation transition, calculable once ξ and the transition dynamics are known. A precise estimate requires numerical simulation of the GPE across the void-to-saturation phase transition – a tractable computation that we flag as the primary numerical prediction of this paper.

3 The Resonant Cosmos: Universe as a Single Eigenmode

3.1 Synthesis

The five stages of Section 2, taken together with the results of Papers I–IV, yield a complete and self-consistent picture of the universe:

Level	Physical Object	SSV Identification
Vacuum	Saturated superfluid at ρ_0	Ground state of GPE
Particles	Topological defects	Vortex rings & breathers
Forces	Pressure-gradient interactions	EM, strong, weak, gravity
Time	Irreversible wake evolution	Entropy = turbulence
Galaxies	Coherent planar solitons	BH-pinned standing waves
Universe	Single resonant eigenmode	Global coherent state of Ψ

3.2 The Universe as Self-Tuned Instrument

The seed wavelength $\lambda_{\text{seed}} = 2\pi/k$ set in Stage 3 is the fundamental tone of the resonant cosmos. All subsequent structure from the proton Compton wavelength ($\sim 10^{-15}$ m) to the galactic aggregation radius ($r_0 \sim 1.13$ kpc, Paper IV) to the Hubble horizon ($\sim 10^{26}$ m) is a harmonic or overtone of this fundamental scale, mediated by the dimensionless ratios α , α^{-1} , and $G_{\text{eff}}/(c^2\xi)$.

This is not numerology. It follows necessarily from:

1. A single postulate (saturated superfluid);
2. A single initial perturbation (primordial transverse mode);
3. The self-consistent nonlinear dynamics of the GPE.

The universe did not need to be fine-tuned. It was *self-tuning*: the equation of state of the saturated superfluid selects ρ_0 , the healing length ξ , the saturation pressure P_0 , and hence all dimensionless ratios including α from a single internal consistency condition.

3.3 The Cosmological Constant Problem, Resolved

The “problem” is that naive estimates of vacuum energy density exceed the observed dark energy density by $\sim 10^{120}$. In SSV, vacuum energy is the rest-energy of the saturated superfluid at density ρ_0 . It does not gravitate in the conventional sense, because gravity itself is the Bjerknæs force between *perturbations* above ρ_0 , not a response to ρ_0 itself. The dark energy the residual cosmological acceleration corresponds to the slow decrease of ρ_0 as the superfluid gradually relaxes its saturation state at cosmological timescales (a feature already implicit in the logarithmic equation of state of Paper III). The required value $\Lambda \sim 10^{-52} \text{ m}^{-2}$ corresponds to a fractional relaxation rate $\dot{\rho}_0/\rho_0 \sim H_0$ the Hubble constant which is exactly the rate at which the universe’s wake expands into previously unperturbed void regions.

4 Focused Predictions

Two quantitative consequences are especially central to the cosmogonic picture.

- C1. Baryon-to-photon ratio from the void-to-saturation transition.** Numerical GPE simulation of the Kibble–Zurek freeze-out should recover the observed baryon-to-photon ratio to order unity without parameter tuning.
- C2. No separate dark-energy substance.** Late-time acceleration should track the slow relaxation of the saturated background rather than require an independent dark-energy field or cosmological constant sector.

5 Conclusion

Within the scoped SSV programme, this paper carries the cosmological claim alone: the present vacuum is the end state of a staged growth process from permissive non-structure to a saturated, defect-bearing medium.

References

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